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Absolute dating of dedolomitization and the origin of magnetization in the Cambrian Morgan Creek Limestone, central Texas

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ABSTRACT

Goethite associated with dedolomite in the Morgan Creek Limestone in central Texas carries a chemical remanent magnetization (CRM). The pole position (88°N, 96°W; k = 67, α 95 = 2.7), which is close to that of the modern pole, and the fact that only normal directions are present suggest that the component is a post-Brunhes (<700,000 yr) CRM. Dedolomite is common in the Morgan Creek and consists of calcite rhombohedra with relict dolomite. Goethite occurs surrounding the rhombs and in rhombic zones within the crystals. The goethite precipitated when iron was released from ferroan dolomite during the dedolomitization process. The paleomagnetic data, therefore, date the dedolomitization process as modern. Dedolomitization is probably related to surface weathering of the Morgan Creek.

In addition to goethite associated with dedolomite, the Morgan Creek contains hematite replacing fossils and glauconite, and detrital hematite. The Morgan Creek also contains three other magnetic components. One of these components is a viscous remanent magnetization aligned with the present magnetic field. Another component is a Triassic/Jurassic CRM residing in hematite. A fourth component in the Morgan Creek is an early Paleozoic magnetization residing in hematite and minor magnetite and represents the vector addition of a detrital remanent magnetization and one or more CRMs. The results of this study demonstrate that paleomagnetic and petrographic techniques can be used to date a diagenetic event and also illustrate the complex nature of magnetization in a sedimentary unit.

INTRODUCTION

Numerous paleomagnetic investigations of limestones have reported chemical remanent magnetizations (CRMs) acquired long after deposition (for example, McCabe and others, 1983; Johnson and others, 1984). In only a few cases, the specific diagenetic origin of the CRM has been identified. If a specific diagenetic phase could be related to a CRM, then the timing of the diagenetic event which precipitated the phase could be determined. In this study, paleomagnetic and petrographic results are presented that place an absolute date on a specific diagenetic event. The event dated is the replacement of ferroan dolomite by calcite, that is, calcitization or dedolomitization. As a result of differences in crystal lattices, calcite is unable to contain as much ferrous iron as does dolomite. During dedolomitization, therefore, ferrous iron may be ejected and subsequently oxidized to form iron oxides (Evamy, 1963; Katz, 1971). Iron oxides, such as hematite and goethite, frequently carry a stable magnetization associated with their formation. The magnetization carried by hematite and goethite that formed during dedolomitization should, therefore, be related to the timing of that diagenetic event.

The unit investigated is the Cambrian Morgan Creek Limestone Member in central Texas. Chafetz (1972) reported that the Morgan Creek contained iron oxides associated with dedolomite. Based on field evidence, he concluded that the dedolomite was surficial and related to the modern exposure of the Morgan Creek. The Morgan Creek, therefore, should carry a modern component of magnetization related to dedolomite. The primary objective of this study was to isolate the magnetic component carried by the iron oxides associated with dedolomite and to determine if this component has a modern direction.

In addition to the magnetization associated with dedolomite, the Morgan Creek contains several other magnetic components; therefore, this study illustrates the complex nature of magnetization in a sedimentary unit. The study also demonstrates the necessity for careful sampling of sedimentary units and the importance of linking the petrographic and diagenetic properties of a unit to its magnetic history.

GEOLOGIC SETTING AND PREVIOUS WORK

The Cambrian (Franconian) Morgan Creek Limestone is exposed in the Llano uplift region of central Texas. Cambrian rocks in the Llano area were deposited on an irregular Precambrian granite and gneiss surface, which consisted of low domes and shallow basins. The tops of some of the highs reached as high as the base of the Morgan Creek (Barnes and Bell, 1977). The Morgan Creek is a member of the Wilberns Formation (Bridge and others, 1947; Cloud and others, 1945) and was deposited during a major transgressive cycle (Chafetz, 1973). Underlying the Morgan Creek, there is the Welge Sandstone, which represents the initial transgression onto the shelf (Barnes and others, 1963). The Welge grades upward into the lower Morgan Creek (Barnes and Bell, 1977). The Morgan Creek consists mostly of an interfingering limestone unit and a terrigenous unit, and the limestone unit is predominant in the study area (Chafetz, 1973). Deposition of the Morgan Creek occurred in a shallow marine, current-agitated environment, and ooids, fossils, glauconite, and intraclasts are common throughout the unit. The Morgan Creek is gradationally overlain by the Point Peak Shale Member.

According to Morton and Long (1984), deposition in the Llano area was relatively continuous from the Late Cambrian through the Early Ordovician. From the Late Ordovician to the Early Pennsylvanian, the area was intermittently above sea level; a period of deposition in the Early Pennsylvanian followed. During the late Paleozoic, faulting occurred which produced the present structural uplift (Bell and Barnes, 1972). Prior to the advance of the Early Cretaceous sea and the deposition of the Cretaceous units, the Llano area was a low-lying source area (Sheldon, 1979). According to Barnes and others

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(1963), erosion proceeded from the late Mesozoic until the present, forming the modern exposure.

Several studies done on the Morgan Creek have indicated the presence of iron oxides associated with dedolomite (Chafetz, 1972; Barnes and Bell, 1977). In addition to iron oxides associated with dedolomite, Barnes and Bell (1977) reported hematite and limonite replacing glauconite, replacing fossils, coating grains, and as matrix. They suggested several possible periods of iron oxide formation: one during the deposition of the Morgan Creek, one shortly after deposition during a period of local emergence and weathering, and one related to the present weathering cycle.

Previous paleomagnetic work done on the Wilberns Formation consists of studies by Van der Voo and others (1976) and Watts and others (1980). Van der Voo and others (1976) reported a pole at 6.4°N and 158.8°E for the Point Peak Member and suggested that it represents a Cambrian magnetization. Watts and others (1980) reported a pole for the Wilberns Formation at 24°N and 151°E and interpreted this as a Cambrian pole, including it in the construction of their Cambrian loop. They suggested that the magnetization of the Morgan Creek resides almost exclusively in hematite.

SAMPLING AND LABORATORY TECHNIQUES

The sampling area of this study is located ~8.7 km northwest of Johnson City, Texas, along the east bank of the Pedernales River (Fig. 1). This area is 1.5 km north of what is commonly called the "Klett-Walker section" (Cloud and Barnes, 1948; Barnes, 1956). The Cambrian rocks in this area (Fig. 2) were deposited on a Precambrian granitic dome, and the units below the Morgan Creek are thin, if they are present at all (Barnes and Bell, 1977). The Morgan Creek is ~38.5 m thick here and consists of beds from 2 to 46 cm thick that dip at shallow angles to the north.

Approximately 200 standard paleomagnetic cores (2.54 cm diam) were drilled from 19 sites using a portable coring apparatus and were oriented *in situ* using a Brunton compass. The sites were located along 2 km of the eastern bank of the Pedernales River and were distributed within a 20-m interval of stratigraphic section. Samples from single sites were collected from individual beds of one lithology. Sampling was done in areas where field lithology suggested the presence of dedolomite; however, as wide a range of lithologies as possible was collected.

Specimens 2.2 cm in length were cut from the oriented cores for paleomagnetic measurements.

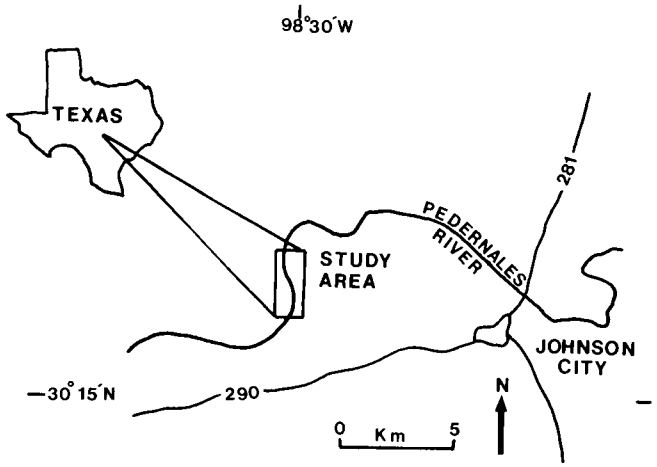


Figure 1. Location of sampling area in Llano region of central Texas.

The specimens were measured using a United Scientific Three-axis Cryogenic Magnetometer. Stepwise alternating-field and thermal demagnetizations were carried out to isolate and identify the components of the natural remanent magnetization (NRM). Alternating-field demagnetization (using a 2-axis tumbling demagnetizer) was not effective in removing a significant portion of the NRM of many of the pilot specimens tested. Thermal demagnetization

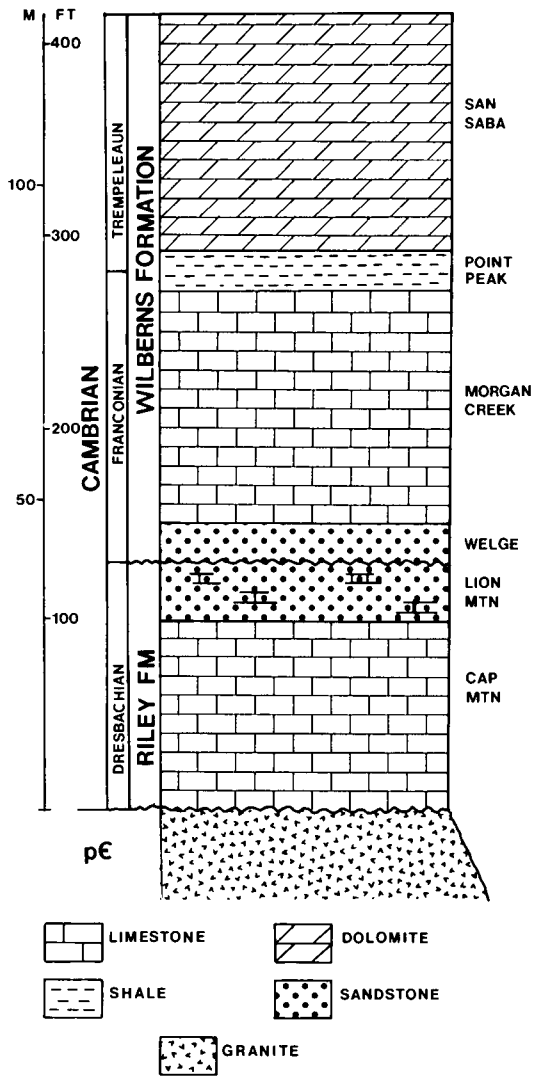


Figure 2. Stratigraphic column of the Upper Cambrian, Klett-Walker section, Llano region, central Texas (after Barnes, 1956).

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was used, therefore, for most specimens. Specimens were heated stepwise from 50 °C to >680 °C in steps of 25 °C or 50 °C in a Schonstedt I Thermal Demagnetizer. During each heating step, specimens were held at constant temperature for 30 min and then cooled to room temperature in a field of <5 nT before measuring the remanent magnetization. Directional and intensity results for each specimen were plotted on orthogonal projections (Zijderveld, 1967) consisting of a vertical and a horizontal projection of the vector endpoints of each demagnetization step. The direction and magnitude of components removed during demagnetization were obtained through visual inspection followed by vector subtraction. Line segments chosen consisted of at least three colinear points. Site mean directions were computed for each site. Mean directions of components were calculated giving equal weight to each specimen. Paleomagnetic poles were calculated from component means.

Coercivity spectrum analysis (Dunlop, 1972) was used to aid in identification of magnetic mineralogies (Lowrie and Heller, 1982). Specimens were subjected in a stepwise fashion to an increasing isothermal remanent magnetization (IRM) approaching strengths of 0.7 T using a water-cooled DC electromagnet. Specimens were then thermally demagnetized at 50 °C intervals to 700 °C.

Chips cut from the paleomagnetic cores were used for the petrographic analysis. Thin-section analysis using transmitted and reflected light was conducted to identify major mineralogic components, such as iron oxides, dolomite, calcite, and opaque minerals, and to determine other diagenetic features. In addition, samples were examined using a scanning electron microscope (SEM) that has energy dispersive capability.

TABLE 1. SITE-MEAN PALEOMAGNETIC DATA

Group: site	N/No	Before structural correction Decl/Incl (degrees)	k	α95	After structural correction Decl/Incl (degrees)
1:E	7/ 7	95/ 22	5.9	27.1	95/ 23
G	9/ 9	106/ 5	17.3	12.7	106/ 7
I	3/10	123/ 6	10.4	40.4	122/ 16
J	6/ 7	106/- 1	11.1	21.0	106/- 1
K	7/ 7	116/ 17	5.5	28.4	116/ 17
S	8/ 9	83/ 15	18.1	13.2	82/ 16
T	5/11	102/ 9	4.8	39.1	100/ 17
U	6/12	115/-14	6.7	28.0	116/- 4
V	5/10	110/ 6	5.4	36.2	110/ 15
2:I	8/10	353/ 48	120.4	5.1	349/ 37
T	8/11	0/ 55	55.4	7.5	354/ 45
U	12/12	3/ 51	103.9	4.3	357/ 41
V	3/10	348/ 46	83.9	13.5	345/ 36
W	2/ 3	(356/ 51)	—	—	(351/ 41)
X	6/ 7	9/ 56	38.2	11.0	1/ 47
Y	3/ 3	7/ 51	418.6	5.5	0/ 42
3:A	7/10	10/ 57	33.3	10.6	2/ 54
N	6/10	18/ 60	20.6	15.1	14/ 59
R	9/ 9	10/ 57	29.4	9.6	6/ 54
4:A	2/10	3/ 9	—	—	4/ 13
B	10/10	3/ 4	14.2	13.3	3/ 6
N	4/10	357/ 10	18.4	22.0	18/ 11
T	3/11	12/-18	4.3	68.3	10/-10
U	3/12	353/- 3	18.3	29.7	18/ 15
V	1/10	18/ 19	—	—	22/ 26
X	1/ 7	328/- 7	—	—	328/ 4
C	9/11	229/ 17	41.2	8.1	230/ 17

Note: Decl and Incl are declination and inclination, respectively, in degrees; N/No is the ratio of specimens or sites used in the statistical analysis to the number of samples studied; k is the statistical parameter (Fisher, 1953) associated with the specimen or site mean; $\alpha 95$ is the radius in degrees of the cone of 95% confidence of the mean.

PALEOMAGNETIC RESULTS

Based on results from alternating-field demagnetization, thermal demagnetization, and coercivity spectrum analysis, most specimens from the Morgan Creek Limestone can be divided into four distinct groups (Table 1).

Group One

The first major magnetic component is characterized by a southeasterly and shallow-to-intermediate direction after correcting for the structure (Fig. 3a). All specimens are of reversed polarity. This component occurs both alone and in combination with other components. It occurs as the only component in sites S, E, J, K, and G, and as one of two components in sites I, T, U, and V, appearing after removal of a low-temperature component (group two). IRM acquisition curves suggest the presence of two magnetic phases (Fig. 3b). The steep initial rise below 0.15 T indicates a low-coercivity mineral such as magnetite. The gradual rise above 0.15 T suggests the presence of a high-coercivity mineral such as hematite. Most specimens do not decay during alternating-field demagnetization (Fig. 3c), although some specimens show significant decay below 60 mT with a southeasterly and shallow-to-intermediate direction (Fig. 3a). In addition, similar directions were removed during subsequent thermal demagnetization of the same specimens (Fig. 3a). Thermal demagnetization of IRM and NRM (Figs. 3d and 3e) does not completely remove the magnetization until temperatures are >600 °C. These high blocking temperatures suggest the presence of hematite.

These results suggest that the predominant magnetic mineral in group one specimens is a high-coercivity mineral (hematite). A low-coercivity mineral (possibly magnetite) is also present at some sites. Directions for specimens in group one are shown in Figure 4, and site mean directions are listed in Table 1. The group mean direction (see Fig. 9 below) for this easterly component has a declination of 104° and inclination of 11°, which corresponds to a pole position of 10.8°N latitude and 158.2°E longitude (Table 2).

Group Two

Group two specimens, which have a northerly and intermediate direction (before structural correction), have NRM intensities from 1.26×10^{-3} A/m to 9.8×10^{-4} A/m. Some specimens contain this component alone, whereas others contain it in combination with the group one component (Fig. 5a). An IRM acquisition curve gradually increases (Fig. 5b), and saturation is not reached at the highest field strengths available (0.7 T); therefore, magnetization of group two probably resides in a high-coercivity mineral. Alternating-field demagnetization to 100 mT did not significantly affect the NRM (Fig. 5c). Thermal demagnetization of both IRM and NRM (Figs. 5d and 5e) indicates removal of the group two component by 150 °C. The low blocking temperature and high coercivity are characteristic of the mineral goethite (Lowrie and Heller, 1982). In addition, after heating to high temperatures, the yellow-orange color of the specimens changes to a red color. This is similar to that reported by Heller (1978) and is the result of the dehydration and decomposition of goethite to form hematite.

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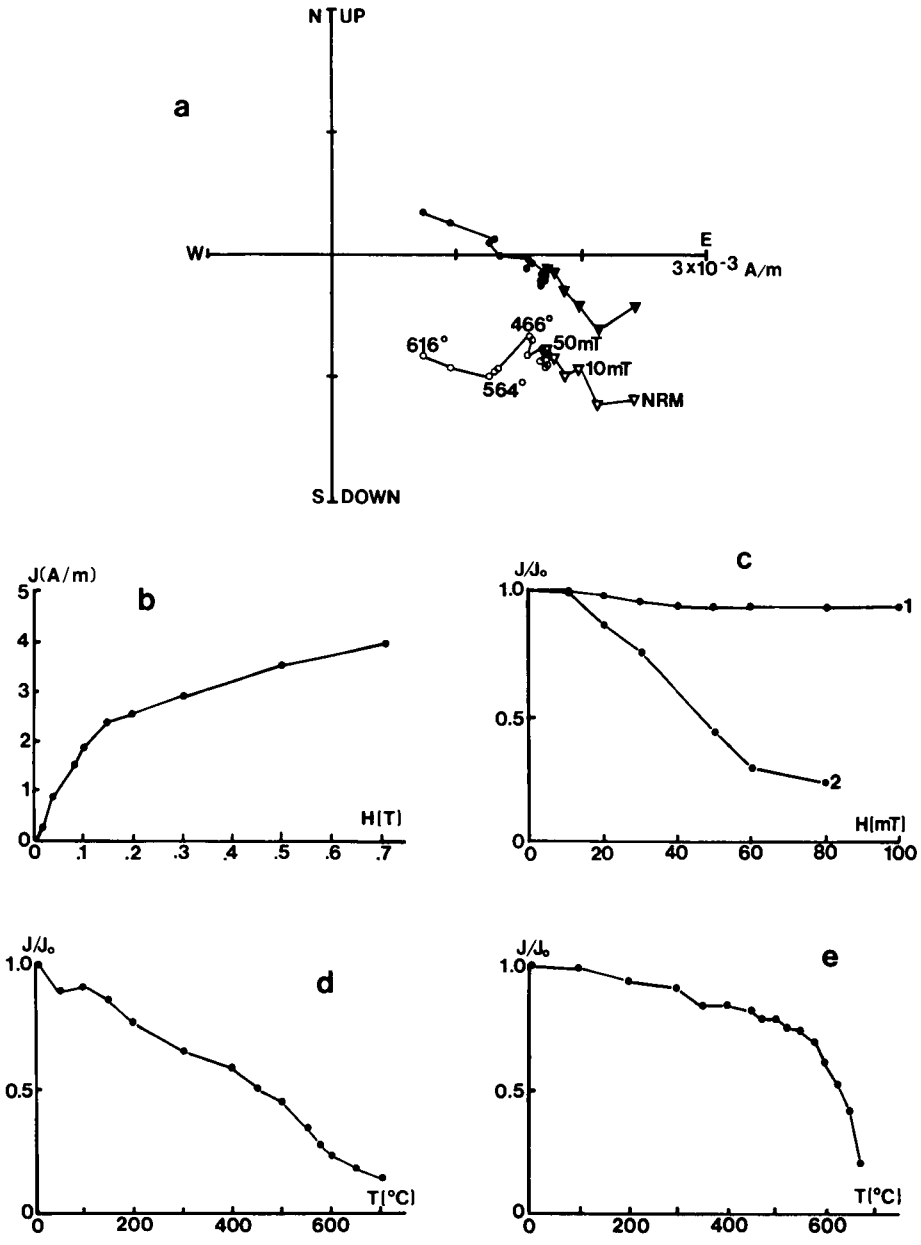


Figure 3. Group one demagnetization and IRM characteristics illustrated by representative specimens. (a) Orthogonal projection of progressive alternating-field (AF) demagnetization (triangles) followed by thermal demagnetization (circles) of specimen E2 which shows the easterly and intermediate direction removed during both thermal and AF treatment (after structural correction). Closed symbols represent projections on the horizontal plane, and open symbols represent projection on the vertical plane. (b) Acquisition of isothermal remanent magnetization (IRM) induced by a direct field (H) on specimen E1b. Note the steep initial increase below 0.15 T and then a more gradual increase at higher field strengths. (c) Stepwise AF demagnetization of NRM (J/Jo is intensity normalized for the initial intensity). Curve 1 (specimen K12) = typical curve for most specimens that show no decay. Curve 2 (specimen G2) = some specimens decay below 60 mT. (d) Stepwise thermal demagnetization of IRM (specimen E1b) indicates that the magnetization is not completely removed until >600 °C. (e) Stepwise thermal demagnetization of NRM (site S mean) which shows that the magnetization is not completely removed by 600 °C.

Directions for the group two component are shown in Figure 4, and the statistics are listed in Tables 1 and 2. Group two consists of only normal polarity directions. The group mean direction has a declination of 0° and an inclination of 51.5° (*see* Fig. 9), which corresponds to a pole position of 87.9°N latitude and 96.0°W longitude.

Group Three

A third component of magnetization found in the Morgan Creek Limestone has a northerly and intermediate-to-steep direction (Fig. 6a). In contrast to the group two specimens, IRM acquisition curves indicate that the magnetization predominantly resides in a low-coercivity mineral (Fig. 6b). The steep initial rise below 0.1 T suggests that the predominant mineral is a low-coercivity phase such as magnetite. The specimens also decay during alternating-field demagnetization (Fig. 6c). The thermal demagnetization of IRM shows an abrupt drop in J/Jo (normalized intensity) at ~575 °C (Fig. 6d). This suggests that the magnetic mineral present is magnetite. Removal of this component during thermal demagnetization of NRM occurs between 200 °C and 400 °C (Fig. 6e).

The specimen directions (Fig. 7) combine to give a group mean declination of 13° and an inclination of 58° (*see* Fig. 9 below; Table 2), which corresponds to a paleomagnetic pole position of 76.5°N latitude and 51.2°W longitude.

Group Four

The fourth magnetic component is northerly and shallow before structural correction (Fig. 8a) and occurs in specimens scattered throughout the sample sites (Table 1). A representative IRM acquisition curve (Fig. 8b) indicates that saturation is not achieved by field strengths of 0.7 T. This component resides, therefore, in a high-coercivity mineral. Alternating-field demagnetization (Fig. 8c) did not significantly affect the NRM. During thermal demagnetization of the NRM and IRM, the magnetization is not completely removed by 600 °C (Figs. 8d and 8e). The magnetic mineral carrying this component is the high-temperature, high-coercivity mineral, hematite.

Directions calculated for specimens containing this component include both normal and reversed directions (Fig. 7). The group mean direction (Fig. 9, Table 2) has a declination of 1° and an inclination of 7°. This represents a paleomagnetic pole position of 63.4°N latitude and 79.6°E longitude.

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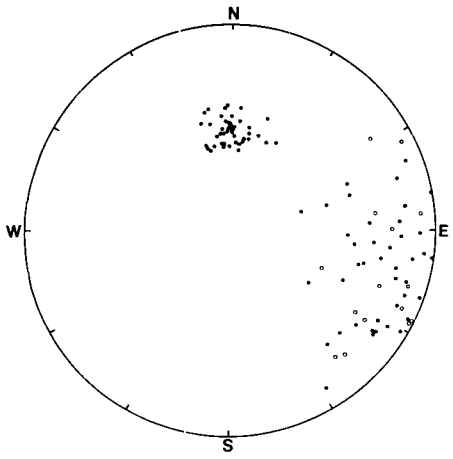
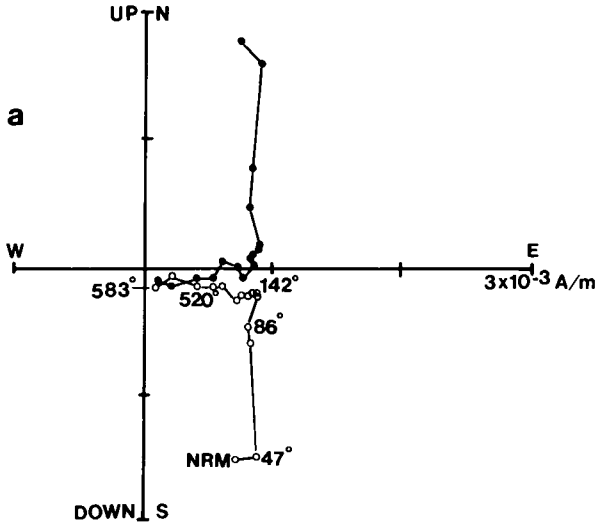


Figure 4. Equal-area plots of specimen directions for group one and group two. Open circles denote negative inclination, and closed circles denote positive inclination. Triangle indicates the modern magnetic direction for the sampling area based on the axial geocentric dipole model.

TABLE 2. GROUP MEAN DIRECTIONS AND POLES

	N/No	Decl/Incl (degrees)	k	$\alpha 95$	Pole position	dp	dm
Group I (after structural correction)							
1a	56/56	104/11	7.3	7.5	9.2°N, 159.4°E*	3.9	7.7
1b	9/9	106/12	29.4	9.7	10.8°N,158.2°E*	5.0	9.9
Group II (before structural correction)							
	42/42	0/51	67.1	2.7	87.9°N, 96.0°W	2.5	3.7
Group III (before structural correction)							
	22/22	13/58	29.6	5.8	76.5°N, 51.2°W	6.0	8.4
Group IV (before structural correction)							
	24/24	1/7	12.3	8.8	63.4°N, 79.6°E	4.4	8.8

Note: Decl and Incl are declination and inclination, respectively, in degrees; N/No is the ratio of specimens or sites used in the statistical analysis to the number of samples studied; $\alpha 95$ is the radius in degrees of the cone of 95% confidence of the mean; k is the statistical parameter (Fisher, 1953) associated with the specimen or site mean; dp and dm are the semi-minor and semi-major axes in degrees, respectively, of the oval of 95% confidence about the poles associated with the paleomagnetic directions; * indicates the pole is reversed 180° from the actual calculated pole position. Group mean statistics in 1a were calculated from specimen directions, and all specimens were equally weighted. Group mean statistics for 1b were calculated using site means.



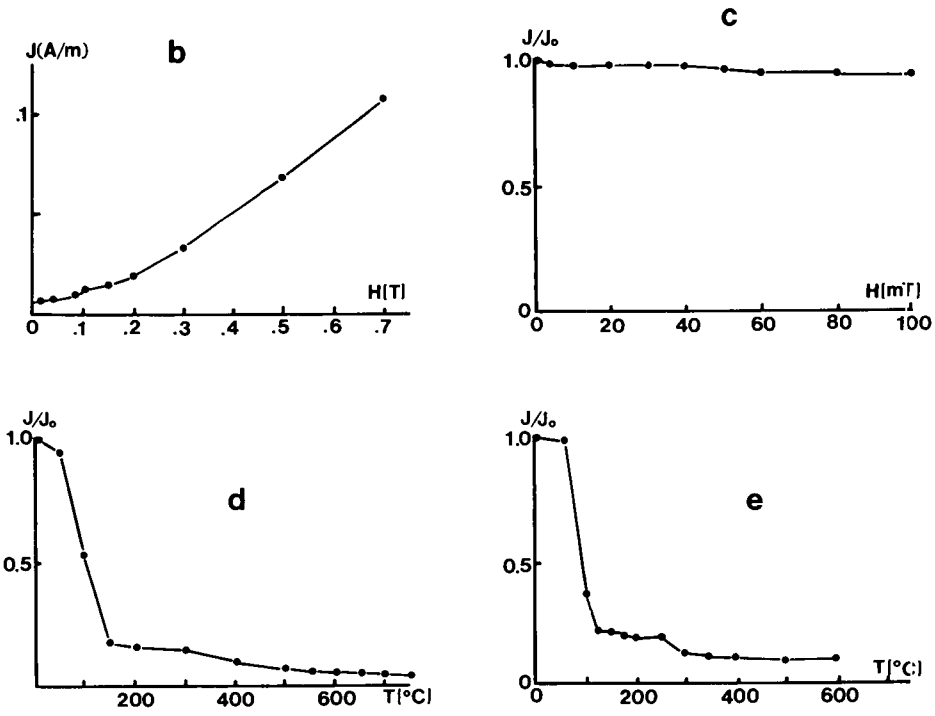


Figure 5. Demagnetization and IRM characteristics for group two. (a) Orthogonal projection (before structural correction) of progressive thermal demagnetization (U6b). The northerly and moderate direction is removed at low temperatures and represents ~80% of the total magnetization. The easterly (group one) direction is removed at higher temperatures. (b) Acquisition of IRM (X1). Intensity increases steadily at fields approaching 0.7 T. (c) Stepwise AF demagnetization of NRM (U3b) which shows no decay. (d) Stepwise thermal demagnetization of IRM (X1). Note removal of the majority of the magnetization by 200 °C, which corresponds to the group two component. The remaining magnetization is not completely removed until temperatures are >600 °C, and it corresponds to the group one component. (e) Stepwise thermal demagnetization of NRM (site Y mean). Results are similar to those of d.

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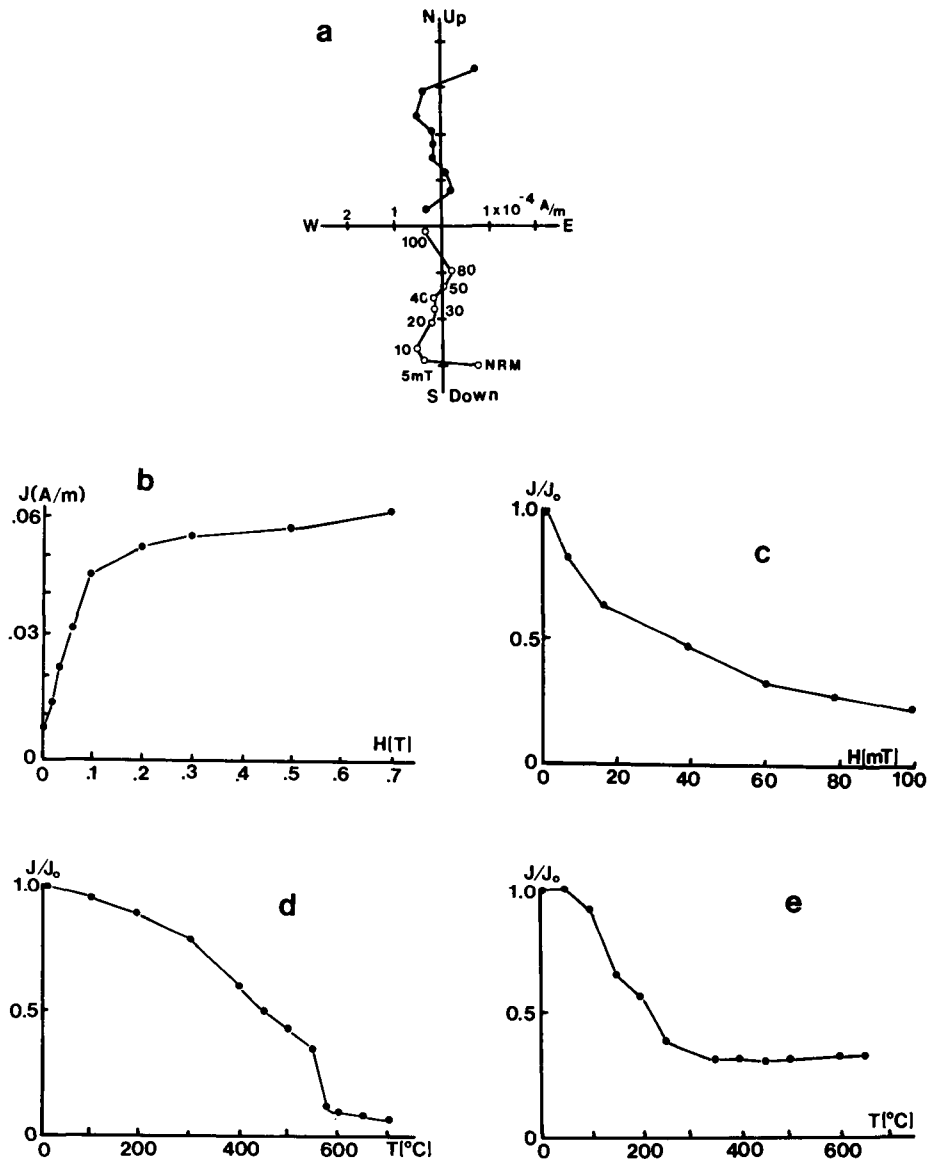


Figure 6. Group three demagnetization and IRM characteristics illustrated by representative specimens. (a) Orthogonal projection (before structural correction) of progressive AF demagnetization (R5a) which shows northerly and steep direction. (b) Acquisition of IRM (A2). (c) Stepwise AF demagnetization of NRM (R5a). (d) Stepwise thermal demagnetization of IRM. Specimen N6 was previously demagnetized using AF demagnetization which removed the group three component. (e) Stepwise thermal demagnetization of NRM (A14b). The break in slope at ~300 °C probably corresponds to the removal of the group three component.

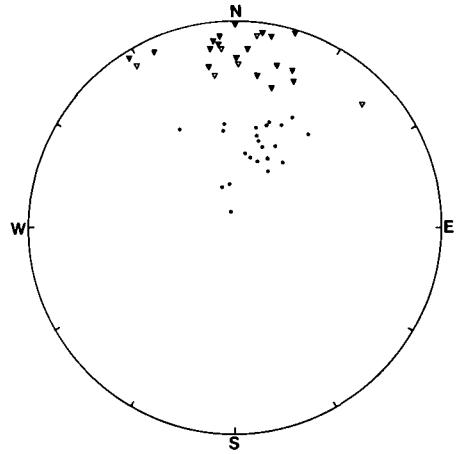


Figure 7. Equal-area plot of specimen directions of group three and group four. Reversed group four directions (2 specimens) are plotted as their antipodal equivalents. Circles represent group three specimens, and triangles represent group four specimens.

Specimens from three of the four remaining sites demagnetized in random and erratic manners and had indeterminable directions and no stable endpoints. The magnetization in specimens from these sites is probably the vector sum of two or more unresolved components. A fourth site, site C (Table 1), did exhibit a consistent site direction. Because the pole position from this site is far removed from the Phanerozoic Apparent Polar Wander Path (APWP), it also appears to be a vector sum of unresolved components.

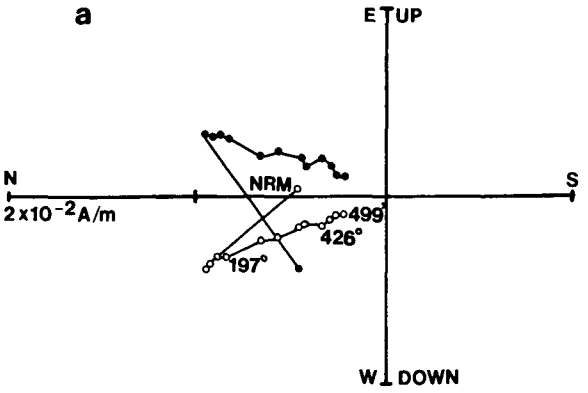
LITHOLOGIC AND PETROGRAPHIC RESULTS

The Morgan Creek Limestone varies in composition from a biomicrite to biosparite and oosparite. Almost all specimens contain fossils (brachiopods, trilobites, and crinoids) and rounded and lobate glauconite grains. In addition, all sites in the Morgan Creek appear to have been dolomitized to some extent.

In many of the sites (I, T, U, V, W, X, Y), the dolomite rhombs have been at least partially replaced by calcite, or dedolomitized. These dedolomite crystals appear in thin section as well-developed crystals that contain rhombic zones of iron oxides (Figs. 10A and 10B). Thin sections stained with Alizarin Red-S show calcite and relict dolomite zones present within the rhombs. Examination using the SEM shows that the dedolomite rhombs typically consist of thin rhombic zones of relict dolomite that alternate with zones of calcite and iron oxides (Fig. 10C). Examination of the crystal morphology of the iron minerals reveals them to consist of bundles and radiating aggregates of prismatic crystals typically 1 μm in length (Fig. 10D). This is the characteristic habit of goethite (Frank, 1981).

In addition to occurring within dedolomite rhombs, iron oxides are present in other forms within the Morgan Creek. Hematite is common as coatings around fossils, replacing fossils, and filling pores within crinoid fragments (Figs. 11A and 11B). This hematite associated with the fossils is concentrated in and around the fossils and does not occur in syntaxial overgrowths (Fig. 11B). This suggests that the hematite was introduced prior to lithification of the unit. The association of hematite with fossils is similar to that reported for Clinton-type iron formation. As in the case of many Clinton-type iron-formations, the hematite was probably introduced during an early stage of diagenesis (Kimberley, 1979). Hematite after glauconite is also common in some of the samples, as evidenced by gradational rims of hematite around glauconite grains and iron-oxide staining (Fig. 11C). Morton and Long (1984) suggested that the periods of glauconite diagenesis occurred during the Late Ordovician and the Early Mississippian.

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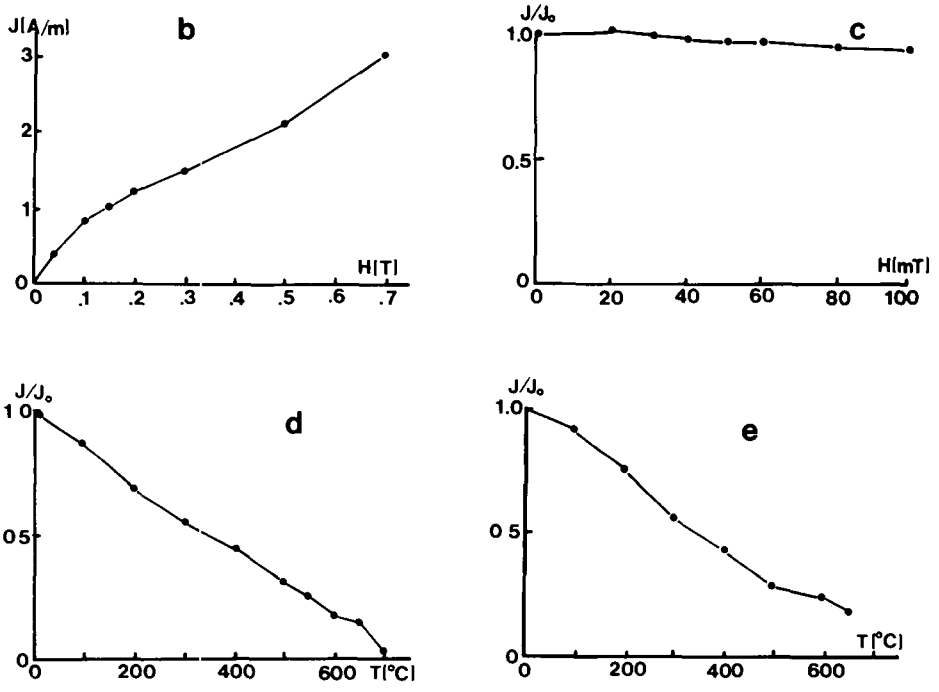


Figure 8. Group four characteristics illustrated by representative specimens. (a) Orthogonal projection (before structural correction) of progressive thermal demagnetization (V9b) which shows northerly and shallow direction. (b) Acquisition of IRM (V4a). Intensity increases at field strengths approaching 0.7 T. (c) Stepwise AF demagnetization of NRM (T1a) indicates no decay <100 mT. (d) Stepwise thermal demagnetization of IRM (V4a). Group four component is not completely removed until temperatures are >600 °C. (e) Stepwise thermal demagnetization of NRM (site N mean); results similar to those of d.

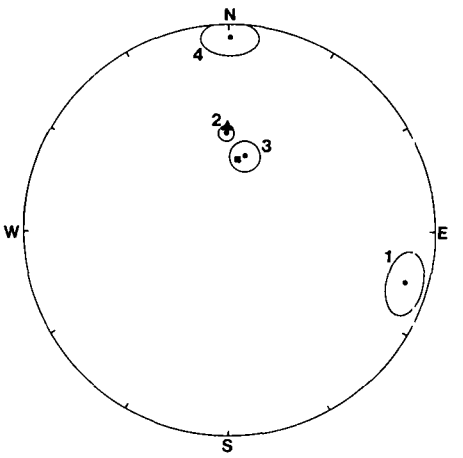


Figure 9. Equal-area projection of group means and cones of 95% confidence for group one (1), group two (2), group three (3), and group four (4). The square indicates the present-day field at the sampling locality; the triangle corresponds to the modern direction based on the axial geocentric dipole model.

All sites in the Morgan Creek contain opaque minerals (Fig. 11D), and the most abundant is hematite. The majority of the opaque hematite is detrital, but, in some cases, hematite appears to be pseudomorphic after pyrite. Hematite also occurs along stylolites and as concentric bands within ooids.

A comparison of the petrographic results with the paleomagnetic results shows that the specimens and sites containing goethite associated with dedolomite make up group two, which coincides with the results for coercivity spectra analysis and with the thermal-demagnetization results.

Those sites containing only the easterly component (sites E, S, J, K, and G of group one) contain no dedolomite; however, they do contain various other forms of hematite, such as hematite replacing fossils and glauconite, detrital hematite, and hematite along stylolites. The only forms of hematite found in all group one specimens, however, are hematite replacing fossils and detrital hematite. This matches the coercivity analysis and demagnetization results, which suggests that most of the magnetization resides in hematite.

The specimens of group four, which have the northerly and shallow component, contain several different forms of hematite. This is consistent with the fact that this component always occurs in conjunction with other components. All specimens in group four, however, contain hematite replacing glauconite. This is consistent with the coercivity analysis, which suggests that the magnetization resides in hematite.

The specimens of group three also have no distinct petrographic characteristics. The forms of iron oxides present in group three specimens vary greatly. The only common feature is the presence of opaque minerals in all group three specimens.

INTERPRETATION OF COMPONENTS

Three of the four magnetic components found in the Morgan Creek Limestone can be related to the depositional and diagenetic history of the

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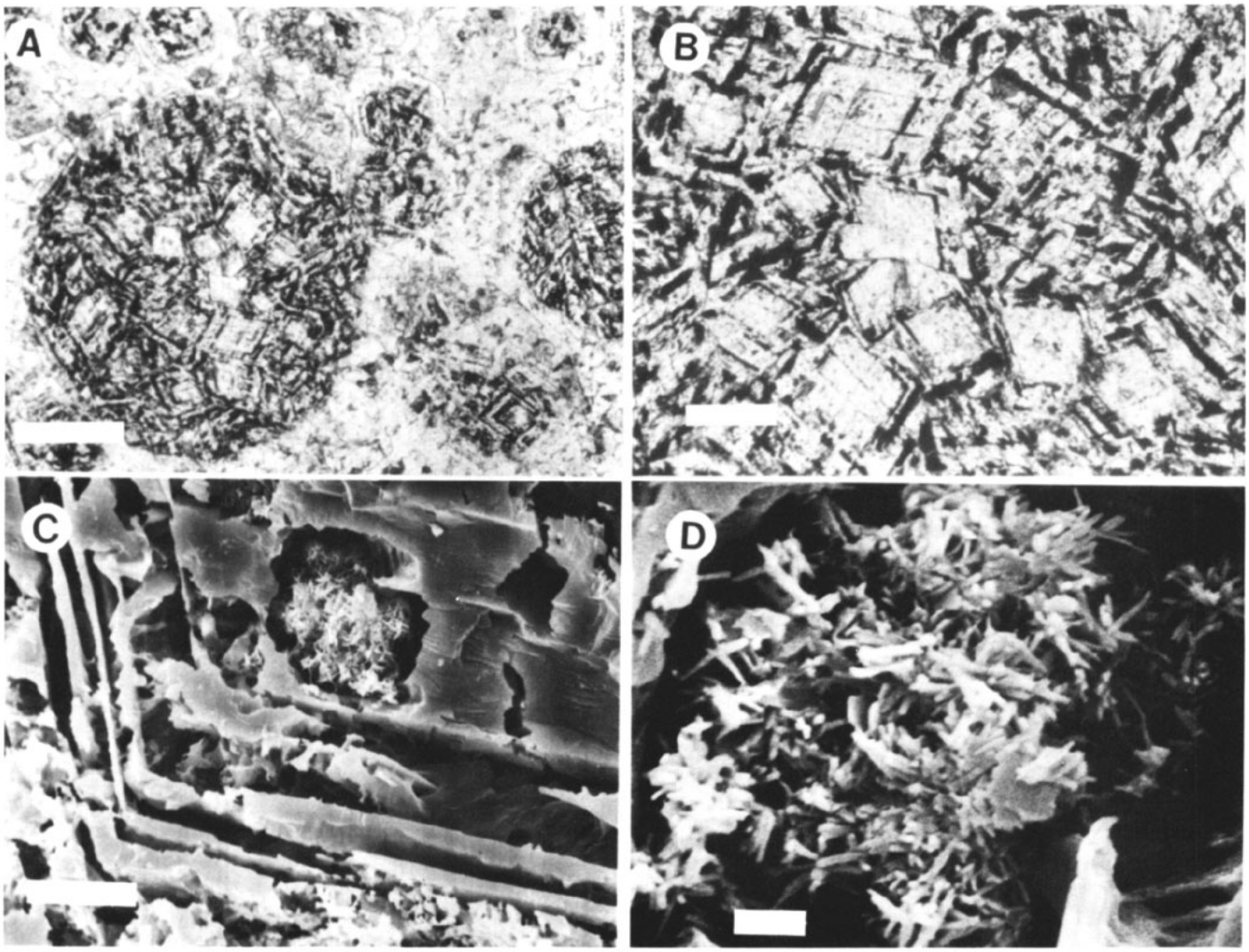


Figure 10. Photomicrographs. (A) Outline of ooid which was replaced by dolomite and then dedolomitized (scale bar = 200 μm). Dark minerals are iron oxides. (B) Dedolomite crystals showing rhombic zones of iron oxides (scale bar = 100 μm). (C) SEM photomicrograph showing outline of dedolomite rhomb, including relict dolomite zones and etched calcite zones. Polished surface, etched for 5–10 sec in dilute hydrochloric acid (scale bar = 10 μm). (D) Goethite within dedolomite crystal (scale bar = 2 μm).

formation, as is discussed below. A fourth component is a viscous remanent magnetization (VRM) which results from the modern magnetic field. A structurally corrected direction was used only for group one because it is the only component which has a magnetization that predates the structure. In any case, because the dips are low (<11°), the correction does not greatly change the directions (*see* Table 1).

The easterly and shallow component (group one) resides in hematite and in minor amounts of magnetite. Group one specimens all contain detrital hematite and hematite replacing fossils, as well as other forms of authigenic hematite. The pole position for this component (10.8°N, 158.2°E) is similar to those for the two previously reported poles for the Wilberns Formation (Fig. 12), which were interpreted by Watts and others (1980) as representing Cambrian magnetizations. The pole position is also similar to an Early Ordovician paleopole at 10°N, 166°E from the Oneota Dolomite (Jackson and Van der Voo, 1985). Because the Cambrian–Early Ordovician part of the APWP is uncertain, all that can be reliably inferred about the age of group one magnetization is that it was acquired in the early Paleozoic.

This component resides in detrital hematite and magnetite as well as in several forms of authigenic hematite. Magnetizations carried by diagenetic magnetite have been reported (McCabe and others, 1983), but the magnetite in the Morgan Creek is probably detrital and should carry a detrital remanent magnetization (DRM). Unfortunately, it was not possible to calculate a direction for this DRM because of the small number of specimens that contain the component. The detrital hematite should also carry a DRM. If the hematite is martite, however, and formed from magnetite during low-temperature oxidation after deposition, then even the hematite grains could carry a secondary magnetization. Other forms of hematite in group one specimens, such as hematite replacing glauconite, pyrite, and fossils, are clearly authigenic and should carry a CRM. This CRM was probably acquired early, as the predominant form of hematite (hematite associated with fossils) ap-

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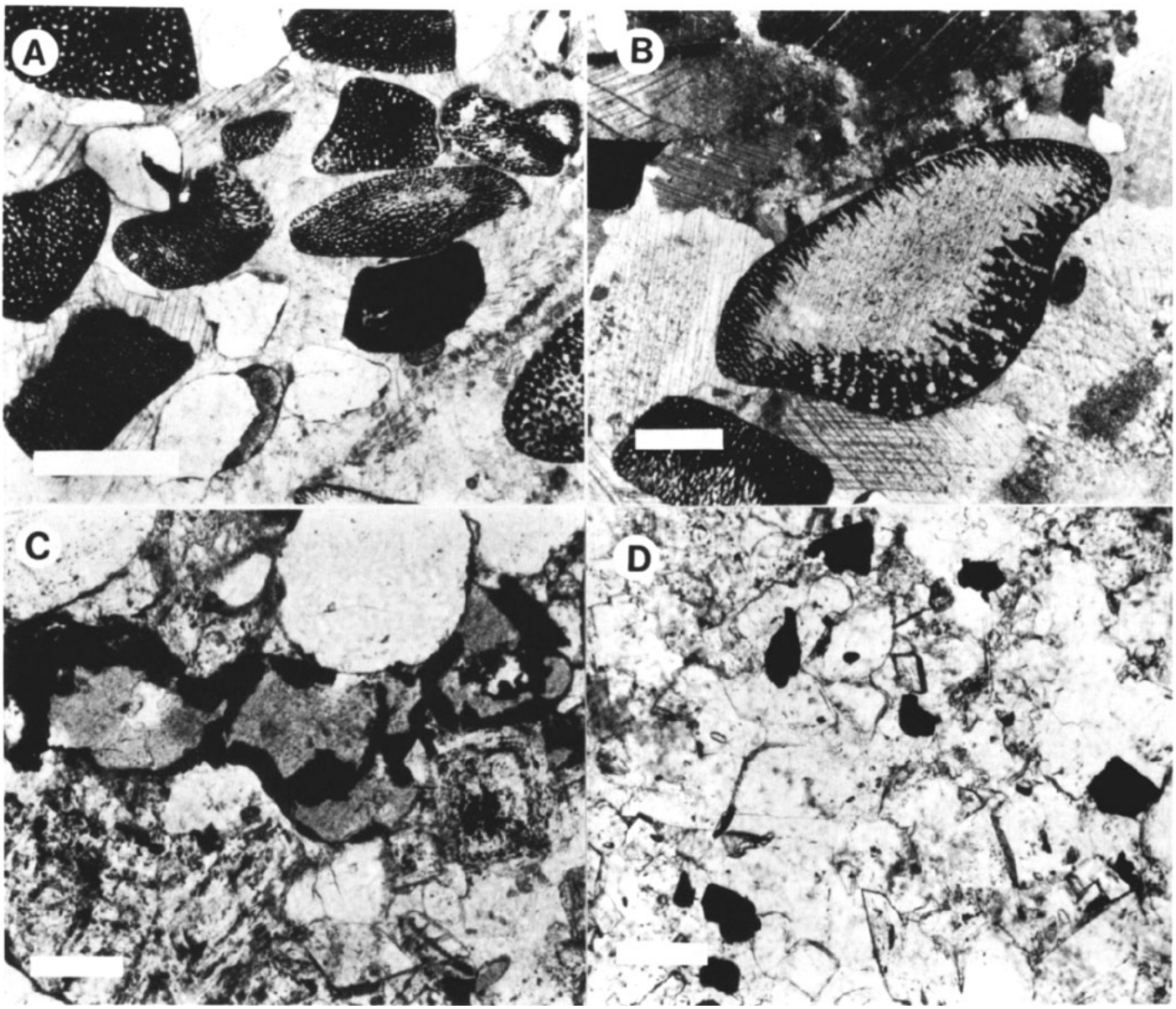


Figure 11. Photomicrographs. (A) Hematite associated with fossil fragments (scale bar = 500 μm). (B) Hematite associated with crinoid fragment. Note absence of hematite in syntaxial overgrowth (scale bar = 100 μm). (C) Hematite replacing glauconite (scale bar = 100 μm). (D) Opaque detrital hematite (scale bar = 100 μm).

parently precipitated during early diagenesis. Because there is more than one form of hematite present in each specimen, the group one component is probably the vector sum of a DRM and one or more CRMs. This component is not, therefore, a primary magnetization.

The pole obtained from group two component (87.9°N, 96.0°W) does not fall near any Paleozoic poles (Fig. 12). The group two magnetization, therefore, was not acquired during the Cambrian and is not a DRM. The Morgan Creek has never been deeply buried or heated to high temperatures; therefore, the possibility of remagnetization by a thermal event can be ruled out. The group two direction and corresponding pole position are similar to the modern direction (axial geocentric dipole field) for the study area (Fig. 9) and its corresponding pole position (Fig. 12). Because this component is resistant to alternating-field demagnetization, it is probably not a VRM. A "hard" VRM (Dunlop and Stirling, 1977) is a possibility, but the petrographic, rock-magnetic, and paleomagnetic data suggest that the component is a CRM. This CRM resides in goethite, and petrographic evidence suggests that the goethite formed during dedolomitization.

The Mesozoic and Cenozoic portion of the APWP for North America proposed by Irving and Irving (1982) is shown in Figure 13. Com-

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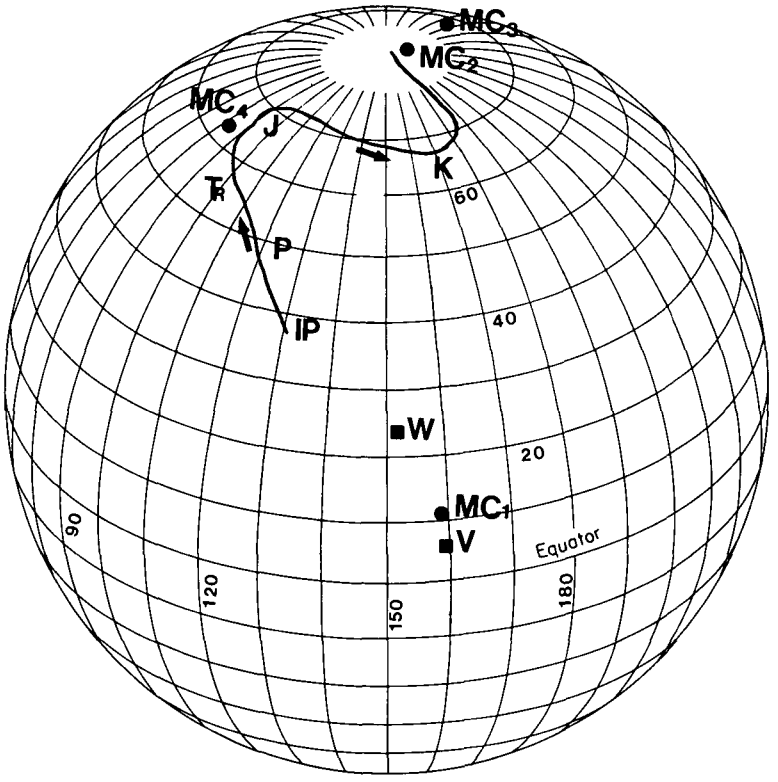
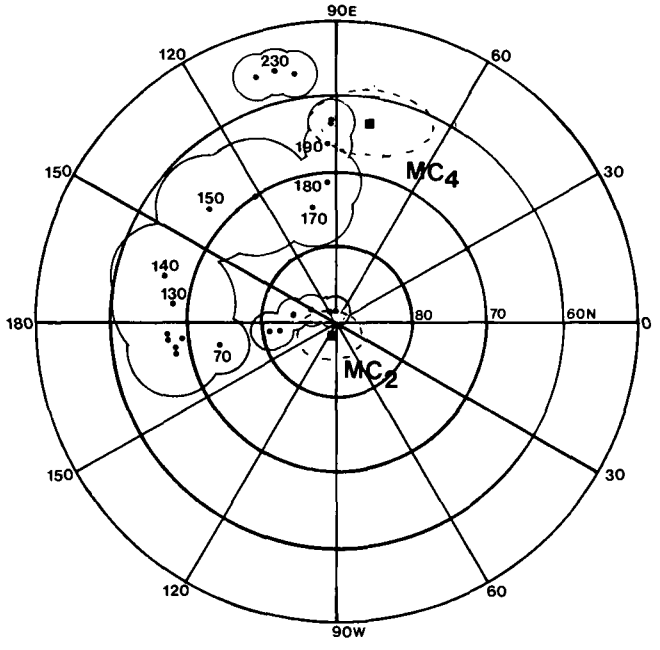


Figure 12. Apparent Polar Wander Path for North America for the late Paleozoic through the present (after Irving and Irving, 1982). Shown are the four Morgan Creek poles of this study (group one = MC1, group two = MC2, group three = MC3, and group four = MC4) and the two previously calculated Wilberns poles (Van der Voo and others, 1976 = V; Watts and others, 1980 = W).



parison of the MC2 pole with this path and the modern pole suggests acquisition of magnetization in late Tertiary to modern time. The overlap of the oval of 95% confidence of the MC2 pole with the modern pole indicates that these two poles are not significantly different at the 95% probability level (McElhinny, 1973). The MC2 pole also falls close to the last 30 Ma of the APWP proposed by Irving and Irving (1982). The absence of any reversed directions within group two suggests that the magnetization which resides in goethite was acquired during one normal polarity event. There have been several periods of normal polarity during the late Tertiary; however, we suggest that goethite precipitated during the modern normal polarity event (past 0.7 m.y.) and is related to modern exposure of the Morgan Creek. The similarity between the group two pole and the modern pole (axial geocentric dipole field) suggests that the magnetization was acquired over the thousands of years necessary to average out changes in the field due to secular variation.

As shown in Figure 9, the direction for the third component in the Morgan Creek is not significantly different at the 95% probability level from the present-day field at the sampling locality. The component is removed during alternating-field demagnetization and has a low coercivity. This component is probably a VRM contained in grains that have low relaxation times and that lost their original primary remanence. Coercivity spectrum analysis of samples from which this component has been removed suggests that these grains are magnetite.

Group four, those specimens containing a northerly and shallow component residing in hematite, has a pole position of 63.4°N latitude and 79.6°E longitude (Fig. 12). The pole falls along the Mesozoic APWP proposed by Irving and Irving (1982) (Fig. 13), which suggests a Late Triassic to Early Jurassic acquisition of magnetization. This magnetization clearly resides in diagenetic hematite and may be related to hematite which formed from glauconite. During the Triassic and Jurassic, the Llano area was a low-lying source area, exposing the Paleozoic rocks to erosion and weathering. This magnetic component is a CRM related to the exposure of the Morgan Creek Limestone.

Figure 13. Apparent Polar Wander Path for North America for the Mesozoic through the present calculated using a 30-m.y. sliding window mean (Irving and Irving, 1982). Circles represent the standard errors. Also shown are the group two (MC2) and group four (MC4) poles of the study. Dashed lines represent ovals of 95% confidence.

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DISCUSSION

Dedolomite can apparently form in a variety of diagenetic environments (*see*, for example, Budai and others, 1983), including both surficial and subsurface settings. Many examples of dedolomite have been related to near-surface conditions (for example, Al-Hashimi and Hemingway, 1973), which has led to the suggestion that dedolomite may indicate an unconformity. Our study confirms that the dedolomite in the Morgan Creek Limestone probably did form as a result of modern surficial weathering.

Limestones commonly have complex magnetizations. This study, as well as others (for example, Johnson and others, 1984), demonstrates that the magnetization and its distribution in limestones can be controlled by the composition and texture of a unit. The composition and texture of limestones are commonly variable, as a result of depositional controls as well as diagenetic changes. Diagenesis plays a particularly important role in controlling the magnetization in limestones. Because of these factors, it is crucial in paleomagnetic investigations that studies of the lithologic-diagenetic features be linked with the paleomagnetic work. As illustrated by our study of the Morgan Creek Limestone, this linkage was necessary to determine the origin of the magnetizations.

CONCLUSIONS

A paleomagnetic and petrographic study of the Morgan Creek Limestone of central Texas has led to the following conclusions.

Based on demagnetization results and coercivity analysis, the Morgan Creek contains four different magnetic components. (a) One is an early Paleozoic magnetization residing in hematite and minor magnetite. This component is probably a vector addition of a DRM and one or more CRMs. The hematite occurs as detrital grains and replacing fossils in addition to other diagenetic forms. (b) The dominant CRM is modern and is contained in goethite, which was formed during dedolomitization. (c) Another CRM, acquired during the Late Triassic to Early Jurassic, resides in diagenetic hematite and may have formed during diagenesis of glauconite. (d) A fourth component present is a VRM aligned with the modern magnetic field.

Paleomagnetic, in conjunction with petrographic, techniques can be used to date and constrain the timing of specific diagenetic events.

Our study demonstrates that the magnetization in a sedimentary unit can be very complex. Diagenetic remagnetization is a major cause of this complexity. Unraveling the complexity commonly requires linking the petrographic and diagenetic properties of a unit to its magnetic history.

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